

Area of Parallelograms

Lesson 5-1

Name: _____

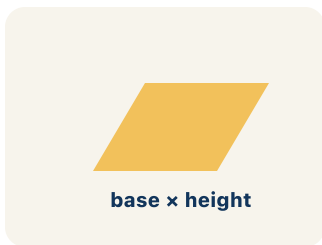
Date: _____

Class: _____

Key Vocabulary

Level 1 support

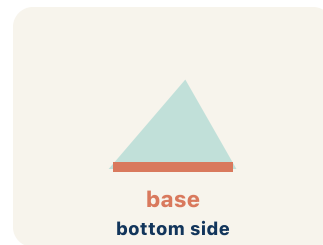
Picture first, then the word, then a plain-language meaning. Say each word out loud.



Think of a leaning door — the top and bottom are parallel, and the two sides are parallel, like a slanted rectangle

Parallelogram

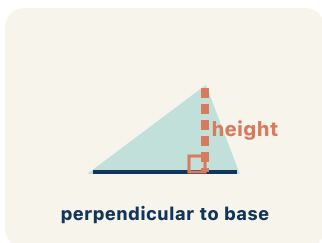
A four-sided shape with two pairs of parallel sides.



If the bottom of a parallelogram is 10 cm, then $b = 10$ cm in the formula $A = b \times h$

Base

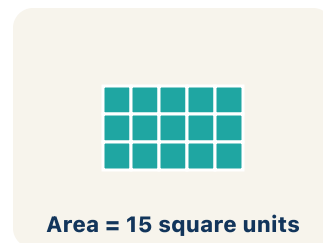
A side of the shape you use to find the area.



A dashed vertical line from the top side straight down to the base, forming a 90° angle — NOT the slanted side

Height

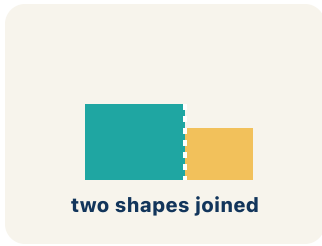
The straight-up distance from the base to the top.



A $3\text{ cm} \times 4\text{ cm}$ rectangle covers 12 square centimeters — imagine 12 tiny 1×1 squares inside it

Area

How much space is inside a flat shape.



An L-shaped room = a 10×8 rectangle joined to a 4×5 rectangle; total area = $80 + 20 = 100$ sq ft

Composite figure

A shape made by putting two or more simple shapes together.

$$3x + 5$$

no equal sign

$A = b \times h$ means Area equals base times height; for a parallelogram with $b = 6$ and $h = 4$, $A = 24$

Formula

A math rule written with symbols.

Key Ideas & Notes



WHAT WE'RE LEARNING TODAY

I can find the area of a parallelogram using $\text{base} \times \text{height}$.

1 Notice

Your architecture firm is designing a parallelogram-shaped patio for a client's backyard.

2 Key idea

I can find the area of a parallelogram using $\text{base} \times \text{height}$.

3 Words I'll use

Parallelogram

Base

Height

Area

Composite figure

Formula



Think About It

- What shape is the patio?
- What measurements are given?
- How is a parallelogram different from a rectangle?

See the notes in action: watch one worked all the way through, then try the next with the same steps.

I do — watch

Follow each step as your teacher solves it.

Problem: What is the area of a parallelogram with base 10 cm and height 7 cm?

- A. 70 sq cm
- B. 34 sq cm
- C. 17 sq cm
- D. 70 cm

Step 1 $A = b \times h = 10 \times 7 = 70$ square centimeters.



Answer: A. 70 sq cm

 We do — together

Solve this one with your class using the same steps.

Problem: A parallelogram has an area of 54 sq ft and a base of 9 ft. What is the height?

- A. 6 ft
- B. 45 ft
- C. 63 ft
- D. 27 ft

Step 1

Step 2

Answer:

 You do — your turn

Now try one on your own. Show every step.

Problem: What is the area of a parallelogram with base 9 cm and height 5 cm?

- A. 45 sq cm
- B. 14 sq cm
- C. 22.5 sq cm
- D. 90 sq cm

Show your work:

Turn & Talk — Discussion Points

Level 1 support

Level 2

Talk with a partner about each prompt. If you need help getting started, use the Level 1 sentence stems. Ready for more? Try the Level 2 question.

LAUNCH

Look at the patio blueprint with a base of 14 feet and a height of 9 feet. Why do we use the height (9 ft) and not the slanted side to find the area?

Level 1 support

Start here: We use the height because it is perpendicular to the base.

 **Need a hint?**

Sentence starters (optional)

We use the height because it is ____ to the base.

Usamos la altura porque es ____ a la base.

The slanted side is not the height because ____.

El lado inclinado no es la altura porque ____.

Word bank: base height area parallelogram formula

Listen for: Students connect 'height' to the perpendicular (straight-up) distance from base to top, and recognize the slanted side is longer and would overstate the area.

Level 2

If the client tilted the patio to make it more slanted but kept the base at 14 ft and the height at 9 ft, would the area change? Justify your answer.

- The area would ____ because the formula only depends on ____ and ____.
- Tilting the shape changes the slanted side but not the ____, so ____.

EXPLORE

On the grid you cut a triangle off one end of the parallelogram and slid it to the other side. What shape did you make, and why does that prove the area is base $\times$ height?

Level 1 support

Start here: Rearranging the parallelogram makes a rectangle with the same base and height.



Need a hint?

Sentence starters (optional)

When I move the triangle, I get a ____.

Cuando muevo el triángulo, obtengo un ____.

Area equals base times ____, so 14 times 9 equals ____.

El área es la base por ____, entonces 14 por 9 es ____.

It is like a rectangle because ____.

Es como un rectángulo porque ____.

Word bank: rearrange rectangle base height area

Listen for: A strong answer names a rectangle, states the area is $14 \times 9 = 126$ sq ft, and explains the rearranged rectangle has the same base and height as the parallelogram.

Level 2

A classmate says any quadrilateral's area is base $\times$ height. Critique this claim using what you know about parallelograms versus other four-sided shapes.

- This claim is wrong because ____ only works when ____.
- A counterexample is ____, which shows that ____.

CONNECT

The landscape architect tiles an 18 ft by 10 ft parallelogram courtyard and each tile covers 2 sq ft. Talk through how you find the number of tiles.

Level 1 support

Start here: First find the area, then divide by the tile size.



Need a hint?

Sentence starters (optional)

First I find the area: 18 times 10 equals ____ square feet.

Primero hallo el área: 18 por 10 es ____ pies cuadrados.

Then I divide by 2 because ____, so she needs ____ tiles.

Luego divido entre 2 porque ____, entonces necesita ____ baldosas.

Word bank: (area) (base) (height) (parallelogram) (tiles)

Listen for: Students compute $\text{area} = 180 \text{ sq ft}$, then divide $180 / 2 = 90$ tiles, and explain why dividing by the tile size gives the count.

Level 2

Tiles come in boxes of 12. Generalize a rule for finding how many full boxes the architect must buy for any courtyard area, and apply it here.

- A rule for boxes is ____ because ____.
- For this courtyard she needs ____ boxes, since ____.

Write About the Math

The Writing Revolution

I can explain how I found the area using the words base, height, area, and formula.

1. Kernel Sentence subject + verb

Model: Parallelogram is a four-sided shape with two pairs of parallel sides.

Paralelogramo es una figura de cuatro lados con dos pares de lados paralelos.

Write a kernel sentence about parallelogram. Use a subject and a verb.

Escribe una oración base sobre paralelogramo. Usa un sujeto y un verbo.

2. Sentence Expansion because · but · so

Kernel: Parallelogram matters in math

Paralelogramo importa en matemáticas

Expand the kernel three ways. Add a reason, a contrast, and a result.

because
porque

Parallelogram matters in math because ____.

Paralelogramo importa en matemáticas porque ____.

but
pero

Parallelogram matters in math, but ____.

Paralelogramo importa en matemáticas, pero ____.

so
entonces

Parallelogram matters in math, so ____.

Paralelogramo importa en matemáticas, entonces ____.

3. Sentence Types 4 ways to write a math idea

Statement
Afirmación

Tell one true fact about parallelogram.
Di un hecho verdadero sobre parallelogram.

Parallelogram ____.

Question
Pregunta

Ask a question about parallelogram.
Haz una pregunta sobre parallelogram.

How does ____ ?

¿Cómo ____ ?

Exclamation
Exclamación

Show excitement about parallelogram.
Muestra entusiasmo sobre parallelogram.

Wow, ____ !

¡Guau, ____ !

Command
Mandato

Tell a partner what to do with parallelogram.
Dile a un compañero qué hacer con parallelogram.

First, ____ .

Primero, ____ .

4. Explain Your Reasoning use a sentence starter

We use the height because it is ____ **to the base.**

Usamos la altura porque es ____ *a la base.*

The slanted side is not the height because ____.

El lado inclinado no es la altura porque ____.

When I move the triangle, I get a ____.

Cuando muevo el triángulo, obtengo un ____.

Try It

Solve on your own. Check the answer key when you are done.

1. Why is a parallelogram's area $\text{base} \times \text{height}$ and not $\text{base} \times \text{slant side}$?

- A. Height is the perpendicular distance between the bases
- B. The slant side is always longer
- C. Both give the same answer
- D. The base is always the longest side

Show your work:

2. A parallelogram has base 12 in and height 4.5 in. What is its area?

- A. 54 in^2
- B. 16.5 in^2
- C. 48 in^2
- D. 27 in^2

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

A parallelogram has a base of 12 cm. If you double the height, the area doubles too. Explain why this happens using the formula, and give a specific example with numbers.

Sentence starter: When the height doubles from ____ to ____, the area changes from ____ to ____ because in the formula $A = b \times h$, the base stays at ____ and ____.

Show your work:

Reflect — Exit Ticket

A parallelogram has a base of 13 inches and a height of 5 inches. What is its area?

- A. 65 sq in
- B. 36 sq in
- C. 18 sq in
- D. 65 in

Your answer:

Answer Key & Teacher Guide

1. **We Try (notes):** A. $6 \text{ ft} - A = b \times h \rightarrow 54 = 9 \times h \rightarrow h = 54 \div 9 = 6 \text{ ft}$.
2. **You Try (notes):** A. $45 \text{ sq cm} - A = b \times h = 9 \times 5 = 45 \text{ square centimeters}$.
3. **Try It 1:** A. Height is the perpendicular distance between the bases — *Area uses the perpendicular height, not the slanted side length*.
4. **Try It 2:** A. $54 \text{ in}^2 - 12 \times 4.5 = 54 \text{ in}^2$.
5. **Exit Ticket:** A. $65 \text{ sq in} - A = b \times h = 13 \times 5 = 65 \text{ square inches}$.

Writing (TWR) — what to look for

- **Kernel sentence:** A complete sentence needs a subject and a verb. Example: Parallelogram is a four-sided shape with two pairs of parallel sides.
- **Expansion:** *because* gives a reason, *but* shows a contrast or exception, *so* shows a result. Answers vary; each must keep the kernel idea and add the correct kind of detail.
- **Sentence types:** Statement ends with a period, question with "?", exclamation with "!", and a command starts with an action verb (a "bossy" verb).